Trophic pyramids

American ecologist Raymond Lindeman proposed a pyramid of energy, called the trophic pyramid, showing the rate at which energy is transferred from one level to the next as herbivores eat plants, and predators eat herbivores. An organism’s trophic level is the position it occupies in a food chain. Plants and algae are at trophic level 1, herbivores at level 2, and the first level of predators is at 3. It is rare for there to be more than five levels. Plants convert the sun’s energy into stored carbon compounds, and when a plant is eaten by a herbivore, some of the energy transfers to the animal. When a predator eats the herbivore, it receives a smaller amount of that energy, and so on.

Published in 1942, Lindeman’s Ten Percent Law explains that when organisms are consumed, only about 10 percent of the energy transferred from them is stored as flesh at the next trophic level. The energy model creates a more realistic picture of the state of an ecosystem. For example, if the biomass of weed and fish in a pond is the same, but the weed reproduces twice as fast as the fish, the weed’s energy would be shown to be twice as large. Also, the pyramid is never inverted—there is always more energy in a lower trophic level than a higher one. Assessing energy transfer, however, requires a lot of information on energy intake, and on the number and mass of organisms.